01-09-2025 – ASE exam

*Student Name and ID:*_____

Question 1

Considering the architecture of a RISC-V processor RV32G with single-precision FP operations presented in the following:

- Integer ALU: 1 clock cycle
- Data memory: 1 clock cycle
- FP multiplier unit: pipelined 8 stages
- FP arithmetic unit: pipelined 4 stages
- FP divider unit: not pipelined unit that requires 8 clock cycles
- branch delay slot: 1 clock cycle, and the branch delay slot disabled
- forwarding enabled
- it is possible to complete instruction EXE stage in an out-of-order fashion.

Using the following code fragment, show the timing of the presented loop-based program and compute how many cycles does this program take to execute?

```
for (i = 0; i < 100; i++) {
    v4[i] = v1[i] + v2[i];
    v5[i] = v2[i] * v3[i];
    v6[i] = v1[i] / v3[i]; }
```

	Clock cycles
V1: .word "100 values"	
V2: .word "100 values"	
V3: .word "100 values"	
V4: .word "100 zeroes"	
…	
.text	
main:	
addi x7, x0, 100	
la x1, V1	
la x2, V2	
la x3, V3	
la x4, V4	
la x5, V5	
la x6, V6	
loop:	
flw f1, 0(x1)	
flw f2, 0(x2)	
fadd.s f4, f1, f2	
flw f3, 0(x3)	

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fsw f4, 0(x4)																																				
fmul.s f5, f2, f3																																				
fsw f5, 0(x5)																																				
fdiv.s f6, f1, f3																																				
fsw f6, 0(x6)																																				
addi x1, x1, 4																																				
addi x2, x2, 4																																				
addi x3, x3, 4																																				
addi x4, x4, 4																																				
addi x5, x5, 4																																				
addi x6, x6, 4																																				
addi x7, x7, -1																																				
bnez x7, loop																																				
nop																																				
TOTAL																																				

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Question 2

Considering the previous program, and in particular the following two instructions:

```
addi  x7, x7, -1
bnez  x7, loop
```

what type of hazard is caused by the use of register x7, and how is it resolved? Justify your answer.

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Question 3

Considering the previous program, and in particular the following two instructions:

```
fdiv.s f6, f1, f3

fsw  f6, 0(x6)
```

what type of hazard is caused by the use of register F6, and how is it resolved? Justify your answer.

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